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Content triage with similarity digests: The M57 case study

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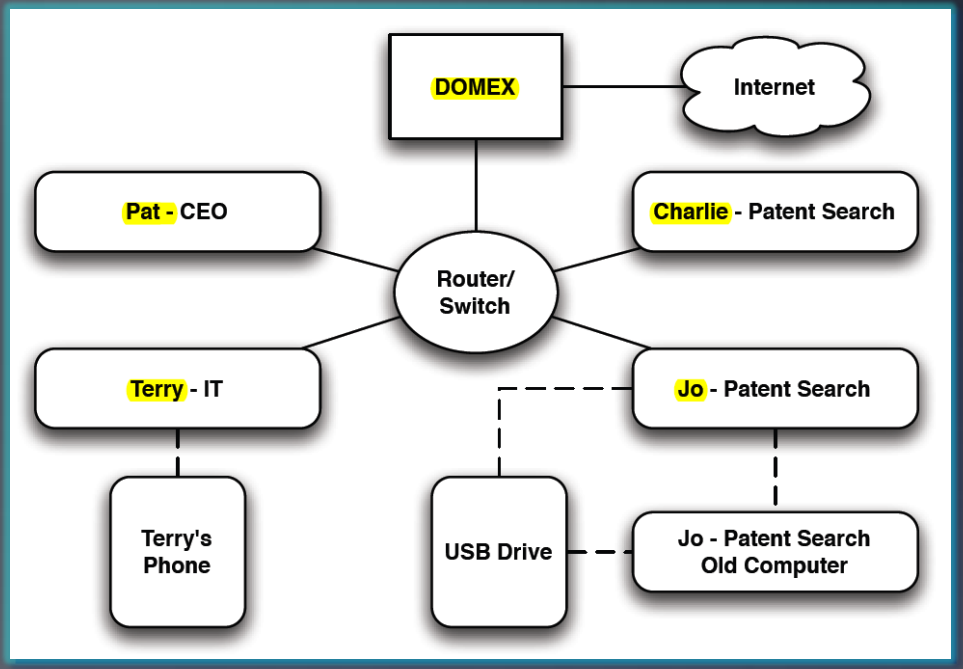
The M57 Case Study

Introduction

M57: The company & setup



- Employees:
  - President: Pat McGoo
  - IT: Terry
  - Researchers: Jo, Charlie
- Period
  - 11/16/2009—12/11/2009
  - 11/20/2009 Jo's computer replaced
  - Last day: police kick down the door
- Data
  - Daily HDD, RAM, network captures

M57: The data (1.5 TB)

- HDD images
  - 84 images, 10-40GB each
  - Total: **1,423 GB**
- RAM snapshots
  - 78 snapshots, 256-1024 MB each
  - Total: **107 GB**
- Network:
  - 49 traces, 4.6 GB
- USB
  - 4.1 GB
- Kitty set
  - 125 JPEGs, 224 MB

Scenario #1: Contraband

- Setup:
  - From the detective reports in the scenario, there is reason to suspect that one of M57's computers (Jo's) has been used in the contraband of "kitty porn".
- Questions:
  - Were any M57 computers used in contraband?
  - If so, when did the accident happen?
  - Is there evidence of intent?
  - How was the content distributed?
  - Was any of the content sent outside the company network?

Scenario #2: Eavesdropping

- Setup:
  - It is suspected that somebody is spying on the CEO (Pat) electronically.
- Plan?
  - Search for potentially rogue processes that might have been introduced on his computer.
  - First HDD image is clean and serves as baseline.

Scenario #3: Corporate espionage

- Setup:
  - There is suspicion that somebody has leaked company secrets.
- Plan?
  - Search RAM snapshots for interesting processes

The need for better triage

Triage

- ***Fast, reliable*** initial screen of the acquired data:
  - *fast:* all you can do in 5/10/15/ … min;
  - *reliable:* provides *strong hints* (low FP).
- Goals:
  - Identify the most (ir)relevant targets/artifacts;
  - Build an overall understanding of the case—what are the likely answers?
- Location of work:
  - We assume post-acquisition work in a lab, but
  - It could be done in the field (given enough hardware)

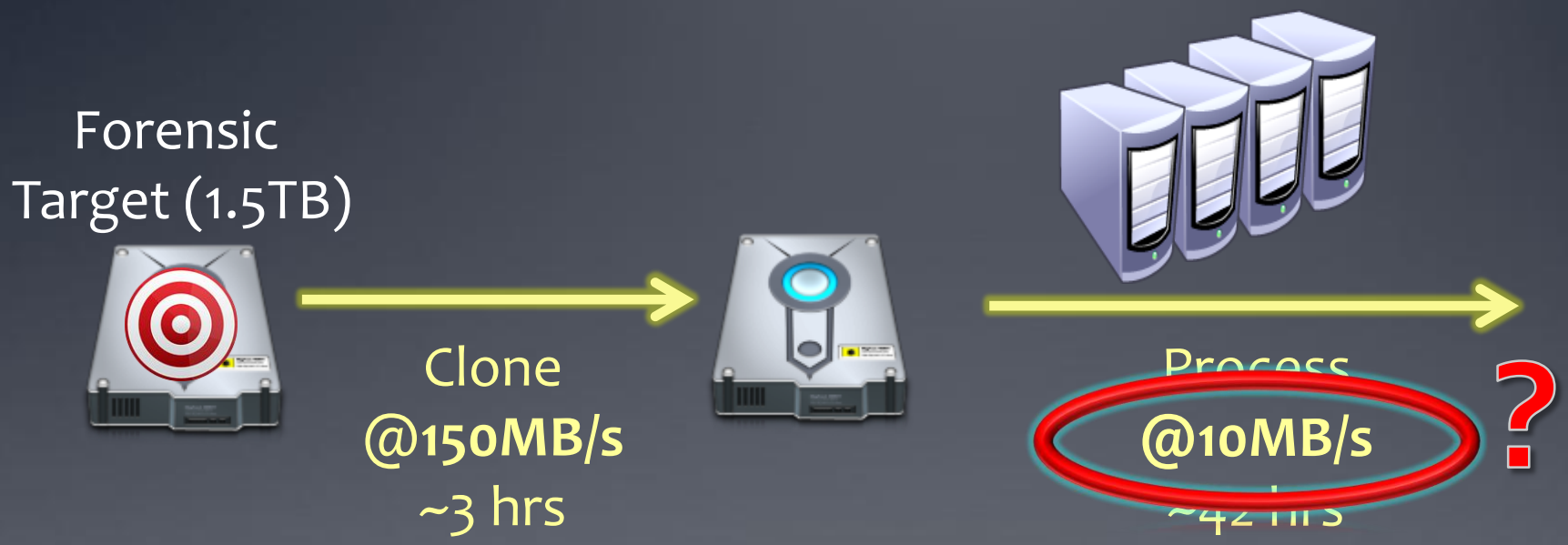
Metadata- vs content-based analysis

- Metadata-based analysis
  - Use FS metadata, registry, logs, etc.
  - Pro: small volume, high-level logical information
  - Con: not looking at the data, cannot see remnants, does not work on a data dump (e.g. RAM), metadata is fragile

  ➔ Typical basis for (manual) triage
- Content analysis
  - Works on actual data content
    - Flie/block hashes, indexing, carving, etc.
  - Pro: looking at actual data, can work with pieces
  - Con: large volume, lower level data

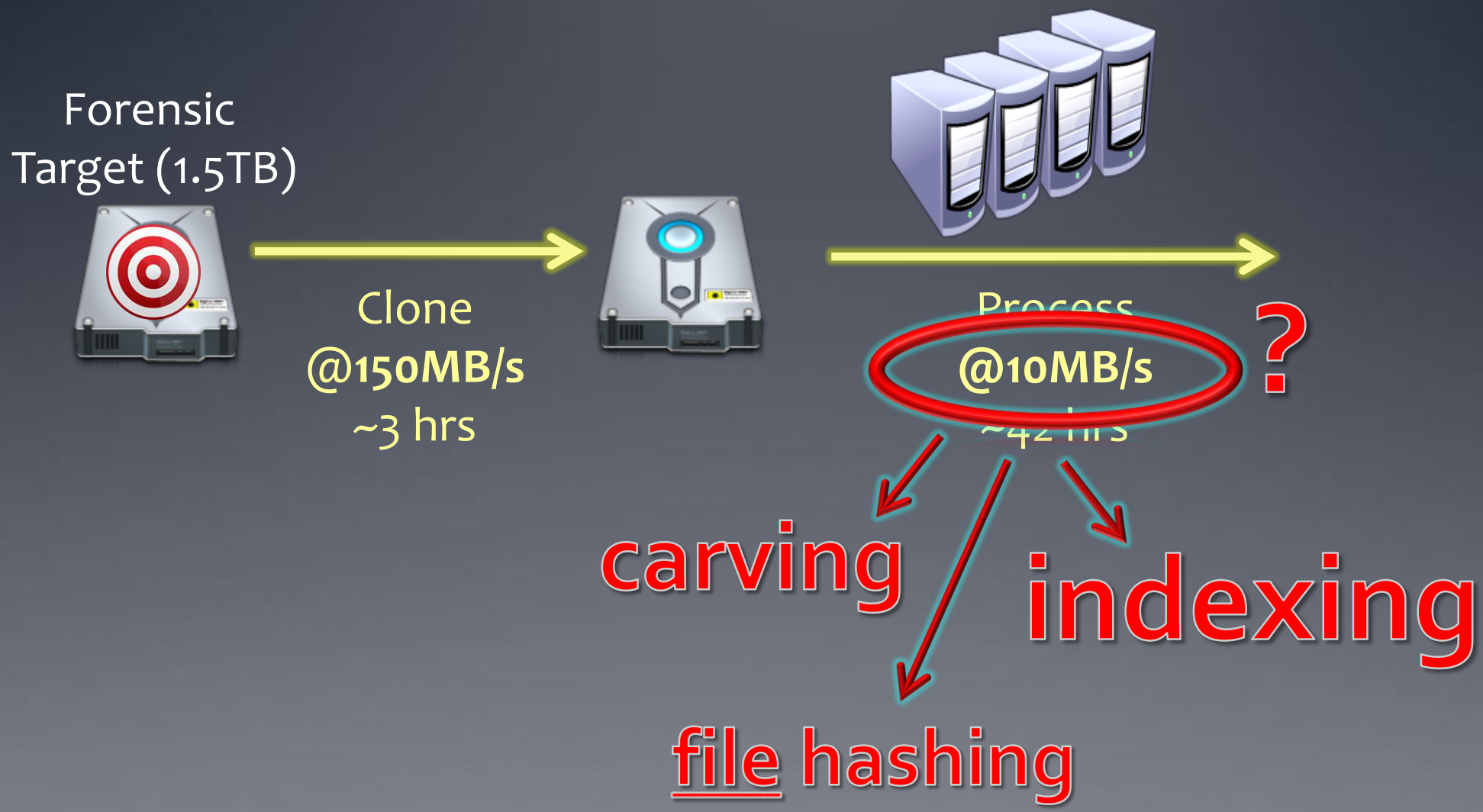
  ➔ Almost never used in triage (perceived as too slow)

Why is content analysis so slow?



➔ We can ***start*** working on the case after 42 hours (!)

Why is content analysis so slow?



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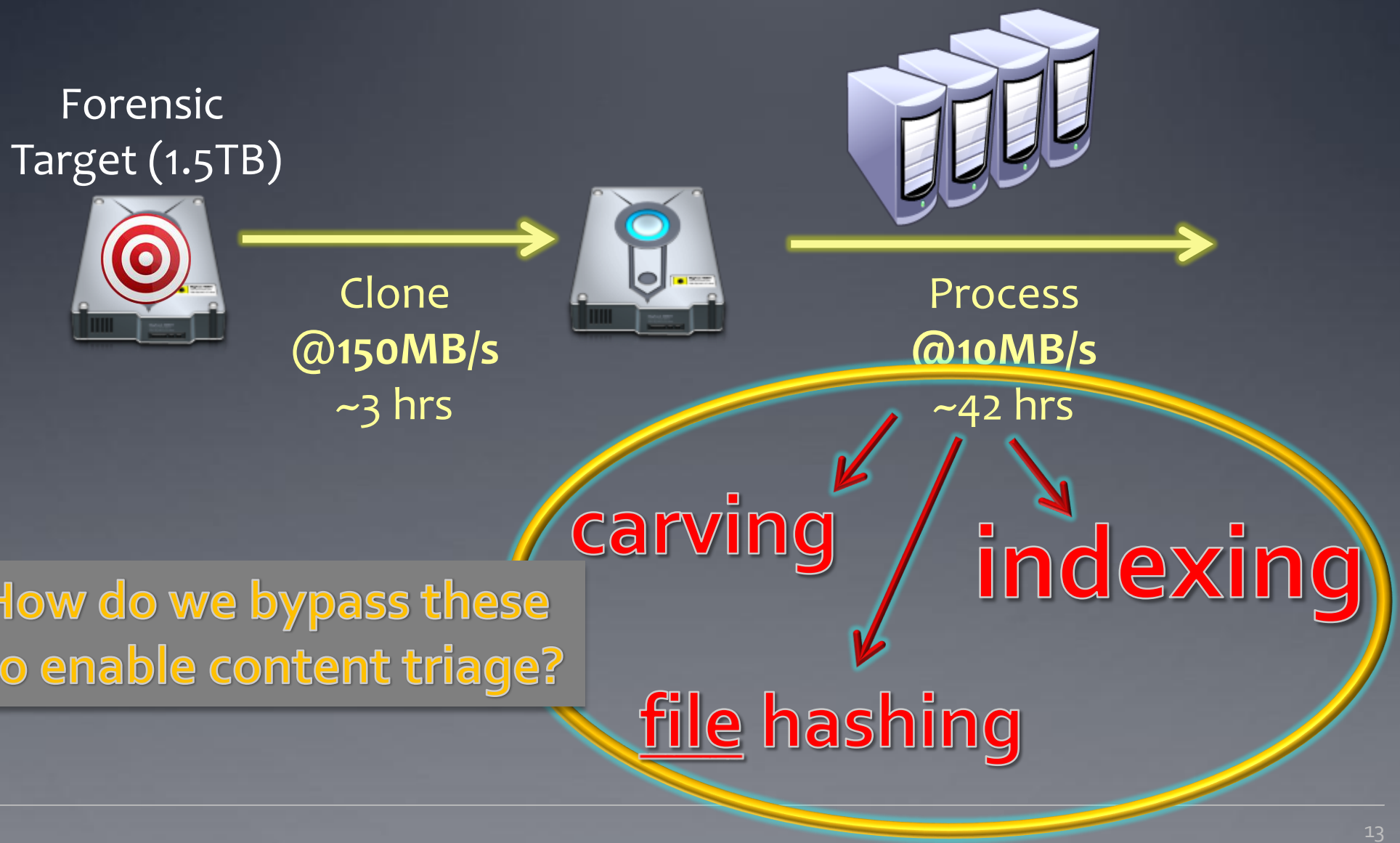


How do we bypass these to enable content triage?

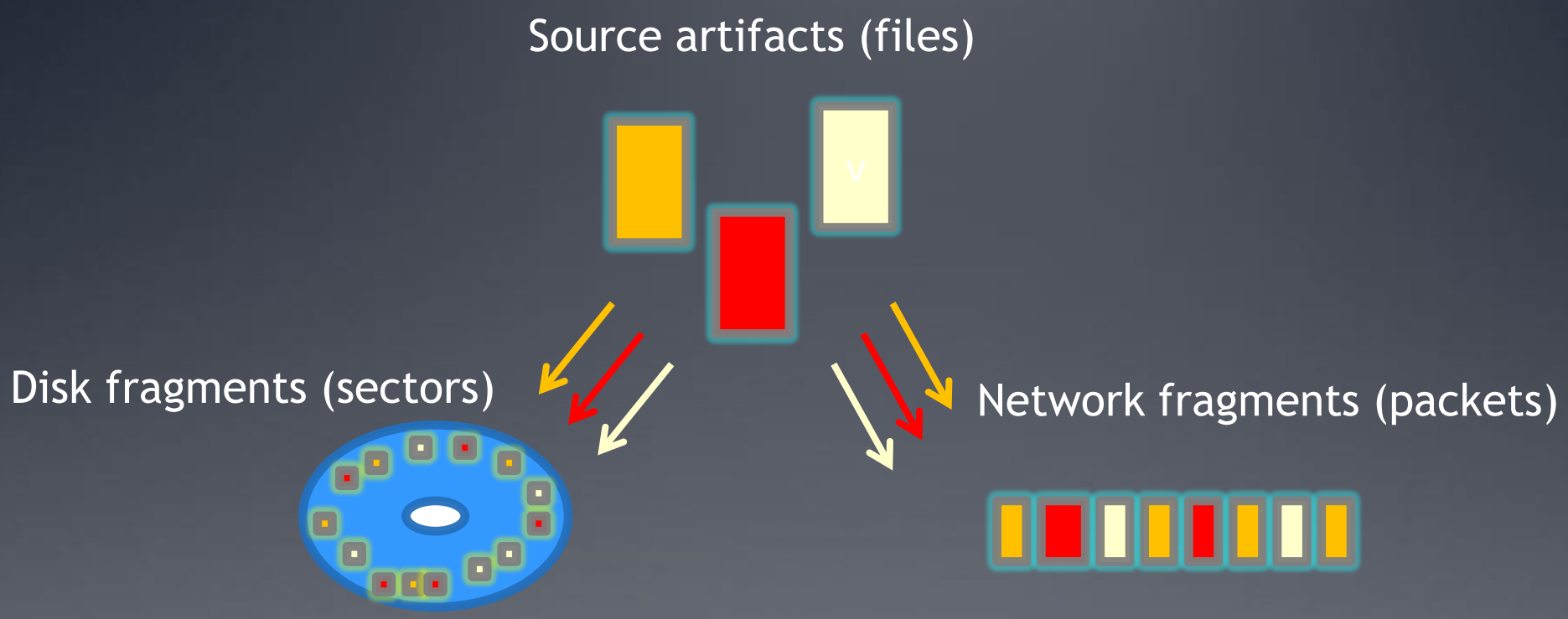
Data Correlation

with similarity digests

Motivation for similarity approach: *Traditional hash filtering is failing*

- *Known file filtering:*
  - Crypto-hash known files, store in library (e.g. NSRL)
  - Hash files on target
  - Filter in/out depending on interest
- Challenges
  - Static libraries are falling behind
    - Dynamic software updates, trivial artifact transformations
    - ➔ We need **version** correlation
  - Need to find embedded objects
    - Block/file in file/volume/network trace
  - Need higher-level correlations
    - Disk-to-RAM
    - Disk-to-network

Scenario #1: fragment identification



- Given a fragment, identify source
  - ***Minimum*** fragments of interest are 1-4KB in size
  - Fragment ***alignment is arbitrary***

Scenario #2: artifact similarity

Bringing science to digital forensics with standardized forensic corpora

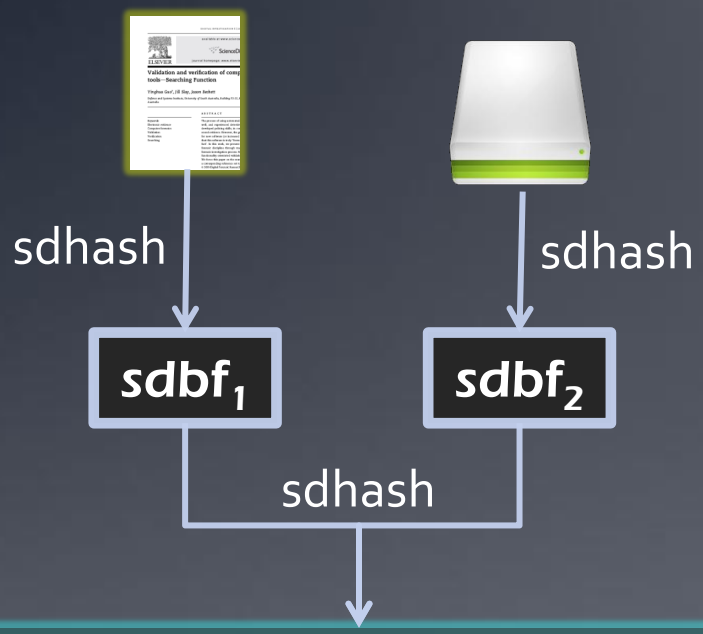
Validation and verification of computer forensic software tools—Searching Function

Similar files
(shared content/format)

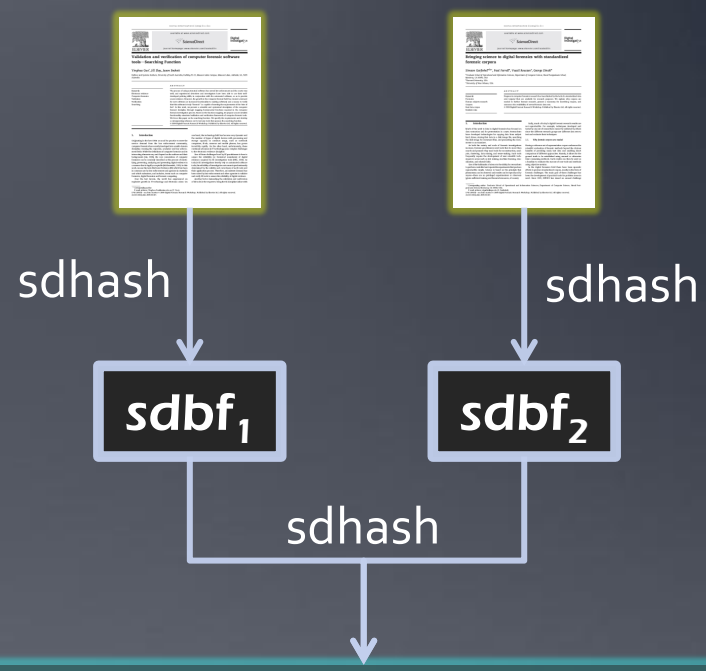
Similar drives
(shared blocks/files)

- Given two binary objects, detect similarity/versioning
  - Similarity here is purely syntactic;
  - Relies on commonality of the binary representations.

Common solution: similarity digests



Is this fragment present on the drive?
➔ 0 .. 100



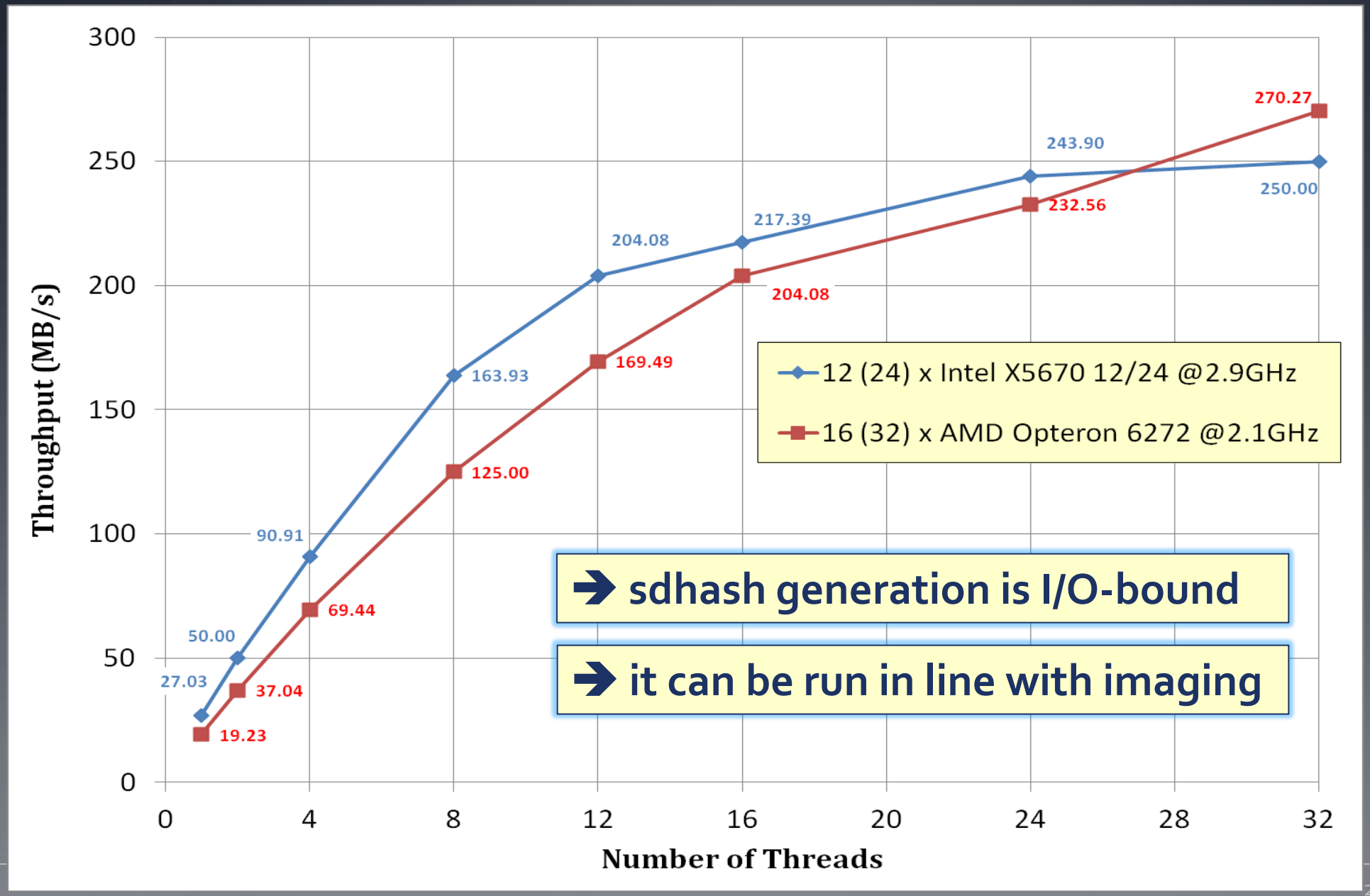
Are these artifacts correlated?
➔ 0 .. 100

All correlations based on bitstream commonality

The M57 Case Study

Using sdhash for triage

sdhash-2.2 generation rates



➔ sdhash generation is I/O-bound

➔ it can be run in line with imaging

sdhash generation times (M57)

Data Set	Size (GB)	Time (min)	Rate (MB/s)
HDD	1,423.0	168.00	143
RAM	107.0	10.70	166
Network	4.6	0.40	196
USB disk	4.1	0.45	155
Kitty	0.2	0.08	45
Total	**1,538.9**	**179.63**	**143**

- Dell PowerEdge R710 server
  - 2 x Intel Xeon CPUs @2.93GHz six-core with H/T 12(24) threads
  - 72GiB of RAM @800MHz

Scenario #1: Contraband

- Setup:
  - From the detective reports in the scenario, there is reason to suspect that one of M57's computers (Jo's) has been used in the contraband of "kitty porn".
- Questions:
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  - If so, when did the accident happen?
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  - How was the content distributed?
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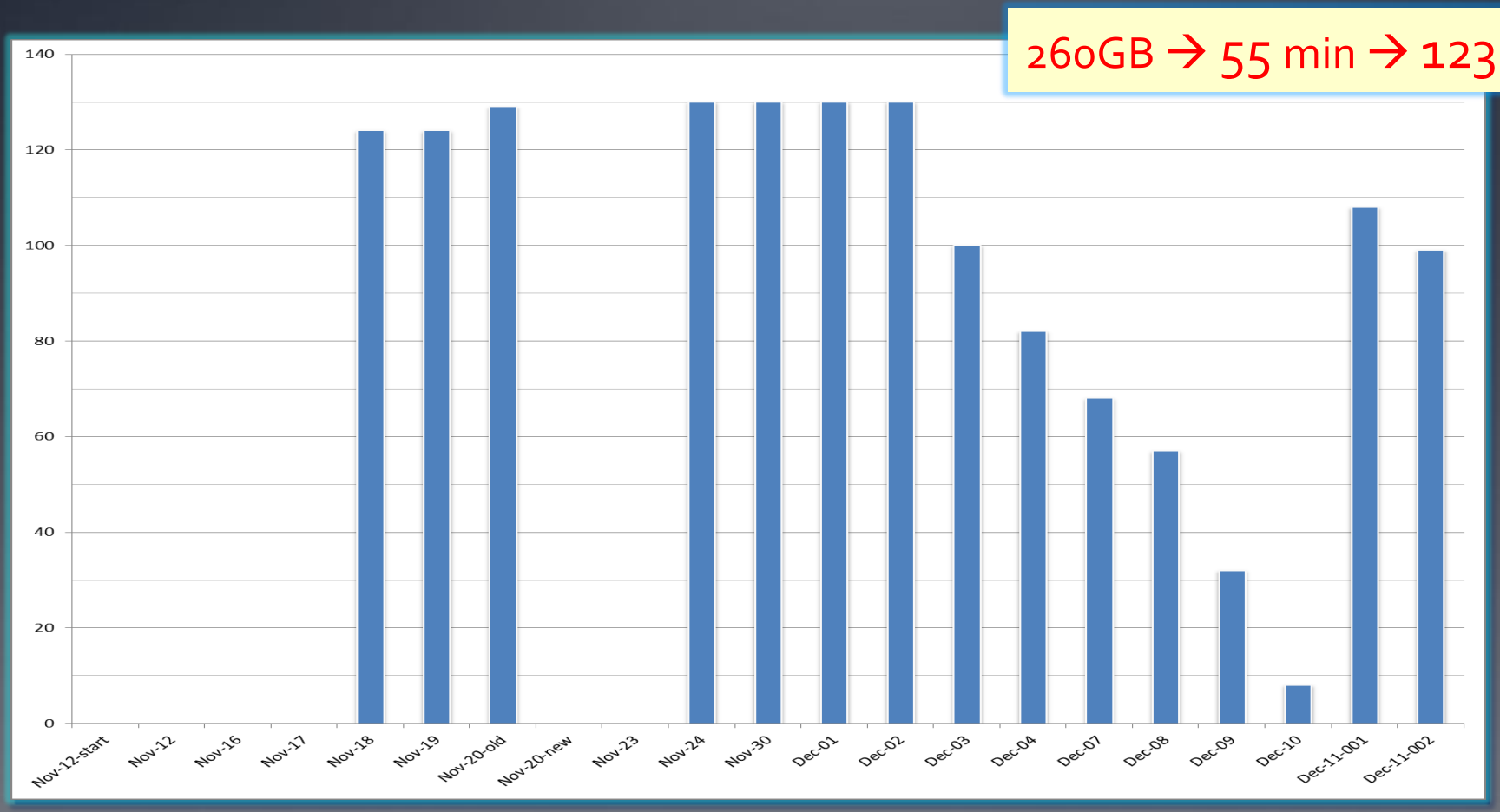
Query 1: Search Jo's HDD for kitty images



Jo's computer: Number of instances found by date

Query 2: What processes were running?

- Search Jo's RAM for traces of installed executables

18 min

```
12/03
.../Downloads/TrueCrypt Setup 6.3a.exe 092
.../TrueCrypt Format.exe               090
.../TrueCrypt Setup.exe                092
.../TrueCrypt.exe                      092
12/04
.../Downloads/TrueCrypt Setup 6.3a.exe 063
.../TrueCrypt Setup.exe                063
12/09
.../Downloads/TrueCrypt Setup 6.3a.exe 084
.../TrueCrypt Format.exe               079
.../TrueCrypt Setup.exe                084
.../TrueCrypt.exe                      090
12/10
.../TrueCrypt.exe                      092
12/11 - pre-raid
.../TrueCrypt Format.exe               086
.../TrueCrypt.exe                      079
```

Scenario #2: Eavesdropping

- Setup:
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- Plan?
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Eavesdropping timeline

20 min

```
11/16, [71] not in baseline
  Present: Java, Firefox, python, mdd_1.3.exe.

11/19, [95] not in baseline
  Acrobat Reader 9 installed or updated,
  including Adobe Air.
  18 other programs from 11/16 still present.

11/20, [289]
  Windows Update run: many new dlls in the
  _restore and SoftwareDistribution folders.

11/23, [561]
  Windows Update has run

11/30, [274]
  Likely a Brother printer driver installed.
  Acrobat/Firefox still present.
```

```
12/03, [649]
  AVG has been updated.
  XP Advanced Keylogger appears:
    XP Advanced/DLLs/ToolKeyloggerDLL.dll 087
    XP Advanced/SkinMagic.dll             027
    XP Advanced/ToolKeylogger.exe         024
12/07, [460]
  More Brother printer related files.
  InstallShield leftovers present.
  win32dd present.
  XP Advanced Keylogger is no longer here.
  RealVNC VNC4 has been installed and run:
    RealVNC/VNC4/logmessages.dll 068
    RealVNC/VNC4/winvnc4.exe     046
    RealVNC/VNC4/wm_hooks.dll    023

12/10, [1240]
  AVG updated.
  IE8 and Windows updated.
  VNC still present.

12/11, [634]
  VNC present.
```

Scenario #3: Corporate espionage

- Setup:
  - There is suspicion that somebody has leaked company secrets.
- Plan?
  - Search RAM snapshots for interesting processes

Scenario #3: Findings

31 min

- RAM
  - "Cygnus FREE EDITION" hex editor
    - On 11/24, 11/30, 12/02, 12/03, and 12/10;
  - "Invisible Secrets 2.1"
    - 11/19, 11/20, 11/24, 11/30, and 12/02.
    - `blowfish.dll, jpgcarrier.dll, bmpcarrier.dll`

    ➔ likely stego tool
- USB
  - `insecr2.exe`
  - `/microscope.jpg`
  - `/microscope1.jpg`
  - `/astronaut.jpg`
  - `/astronaut1.jpg`
  - `/Email/Charlie_..._Sent_astronaut1.jpg`
  - `/Email/other/Charlie_..._Sent_microscope1.jpg`

M57 Conclusions

- Using `sdhash`, we can outline the solution of all three cases in about **120 min** of extra processing.
  - This assumes HDD/RAM hash generation while cloning.
  - This could be further improved by running the queries in R/T in parallel with acquisition.
- The tool enables differential analysis that is simple, fast, robust, and generic.
  - ➔ Most processing can run in parallel with acquisition.
  - ➔ In effect, it can replace carving/indexing during triage.
  - ➔ It does not require much expertise to apply; results are intuitive.
  - ➔ The analysis can be highly automated; higher-level analysis can be built on top.